

A Cyclic Polytabloid Proof of Saxl's Conjecture

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Abstract

For every staircase partition, we prove that the tensor square of the corresponding irreducible complex representation of the symmetric group contains every irreducible representation of that group. This proves Saxl's conjecture.

1 Introduction

For a partition $\alpha \vdash n$, let S^α denote the corresponding irreducible complex representation of the symmetric group S_n . The Kronecker coefficient

$$g(\alpha, \beta, \lambda) = \dim \operatorname{Hom}_{S_n}(S^\lambda, S^\alpha \otimes S^\beta)$$

is the multiplicity of S^λ in the tensor product, with S_n acting diagonally. Put

$$N_m = \frac{m(m+1)}{2}, \quad \rho_m = (m, m-1, \dots, 1). \quad (1.1)$$

We prove the following statement.

Theorem 1.1 (Saxl's conjecture). *For every integer $m \geq 1$ and every partition $\lambda \vdash N_m$,*

$$g(\rho_m, \rho_m, \lambda) > 0.$$

Equivalently, $S^{\rho_m} \otimes S^{\rho_m}$ contains every irreducible complex representation of S_{N_m} .

The assertion is uniform in both the staircase size and the constituent. The proof establishes it inside one explicitly generated submodule of the tensor square, a stronger conclusion that retains the vector needed at each induction step.

The companion [Ope26, Theorem 1.1] uses the stronger cyclic statement of Theorem 3.1, together with additional arguments, to construct, for every positive integer $n \notin \{2, 4, 9\}$, an irreducible complex representation of S_n whose tensor square contains every irreducible representation of S_n .

1.1 History and significance

Saxl's conjecture belongs to a broader tensor-square covering problem: can a single irreducible representation have every irreducible as a constituent of its square? A motivating precedent comes from finite simple groups of Lie type. Heide, Saxl, Tiep, and Zalesski proved that the Steinberg square has this property, except for the groups $\operatorname{PSU}_d(q)$ with $d \geq 3$ and $\gcd(d, 2(q+1)) = 1$ [HSTZ13, Theorem 1.2]. Saxl proposed the staircase assertion in a UCLA Combinatorics Seminar on 20 March 2012, as recorded by Pak, Panova, and Vallejo [PPV16, Conjecture 1.2 and Footnote 2]. Their work

places the staircase square among tensor-square covering questions for symmetric groups and proves several families of positive Kronecker coefficients; in particular, their Theorem 1.4 covers hooks and two-row partitions for sufficiently large staircase size. Ikenmeyer proved positivity when λ and ρ_m are comparable in the dominance order [Ike15, Theorem 2.1]. We use his triangular arrangement of two alternating tensor layers [Ike15, Section 4]; his argument also yields all hook constituents [Ike15, Corollary 6.1].

Character-theoretic and modular methods have supplied further substantial families. The character-value criterion of Pak, Panova, and Vallejo [PPV16] detects constituents using the principal-hook cycle type of a self-conjugate partition. Bessenrodt developed related critical-class and spin-character methods, including the double-hook case [Bes18]. Li proved the triple-hook case using computer-assisted finite reductions [Li21]. Here double and triple hooks are diagrams whose largest square has side two and three, respectively. Bessenrodt, Bowman, and Sutton connected Kronecker positivity to 2-modular decomposition numbers; their results include all 2-height-zero constituents, hence all odd-degree constituents, and framed staircases [BBS21, Theorems 5.2 and 5.5].

Another line of work addresses the tensor exponent and asymptotic coverage. Luo and Sellke used the semigroup property to prove that a random partition occurs in the staircase square with probability tending to one, for both the uniform and Plancherel measures [LS17, Theorems 1.5 and 1.6]. They also found, in every sufficiently large degree, an irreducible representation whose fourth tensor power contains every irreducible [LS17, Theorem 1.4]. Harman and Ryba proved full coverage for the tensor cube of every staircase, with combinatorial and modular proofs [HR23, Theorem 1.1]. Asymptotic coverage does not exclude exceptional constituents, while a cube or fourth power uses additional tensor factors; neither gives the square assertion in Theorem 1.1.

Recent developments include semigroup constructions of further constituents and generalized-block constraints on support [Ebr25], and a full staircase-square theorem for unipotent characters of $GL_n(q)$ [LN25, Theorem 1.1]. The latter is an analogue in a different character theory: ordinary Kronecker positivity implies the corresponding unipotent positivity, but the converse need not hold.

The band calculation below also has a precise predecessor at the level of ambient representations. Brown, van Willigenburg, and Zabrocki proved that $S^{(m,m-1)} \otimes S^{(m,m-1)}$ is the multiplicity-free sum of all irreducibles of S_{2m-1} indexed by partitions with at most four rows [BWZ10, Corollary 4.1]. Our induction needs each of these constituents in the cyclic module of one prescribed band vector, rather than merely somewhere in that ambient square. Explicit nonzero pairings supply this additional generator-specific statement. Throughout, the objective is support, or positivity, rather than a formula for Kronecker multiplicities.

1.2 The construction and its reusable steps

Place N_m tensor positions in the triangle $B_m = \{(i, j) : i, j \geq 1, i + j \leq m + 1\}$. Let R_m and C_m be its row and column set partitions, ordered within each row by increasing j and within each column by increasing i . Use the ordered basis $e_1, \dots, e_m$ of $\mathbb{C}^m$. On a block of length r , alternate the initial letters $e_1, \dots, e_r$, taking the signed sum over all permutations. Taking the product over row blocks gives v_{R_m} ; doing the same over column blocks gives v_{C_m} . These are polytabloids of staircase type. Instead of beginning with an arbitrary vector in the tensor square, we work throughout with

$$W_m = \mathbb{C}[S_{B_m}](v_{R_m} \otimes v_{C_m}),$$

where S_{B_m} permutes the positions in both layers simultaneously. Theorem 3.1 asserts that this particular cyclic module already contains every irreducible. We first prove that it contains every S^λ for which λ is dominated by ρ_m : each initial sum of the parts of λ is at most the corresponding staircase

sum. A sign twist and a permutation-module quotient give this starting case (Proposition 3.2). The following three steps reach the remaining partitions.

1. **A full induced quotient from a coordinate projection.** Separate a smaller triangle B_{m-s} from a band of width $s = 1$ or 2 . Call letters $1, \dots, s$ low and letters $s+1, \dots, m$ high. Each position carries one letter from each tensor layer. Project onto words whose two letters at each position are either both high or both low. The staircase row and column counts force the high positions of the projected generator to be exactly the smaller triangle. The generator then factors into the smaller cyclic generator and a band generator, whose cyclic module is denoted by $U_{m,s}$. Translates of the high-position set occupy disjoint coordinate sectors, so the quotient is the full induced module

$$\text{Ind}_{S_{B_{m-s}} \times S_{B_m \setminus B_{m-s}}}^{S_{B_m}} (W_{m-s} \boxtimes U_{m,s}).$$

Lemma 4.1 isolates the general reason that sector separation gives an induced-module isomorphism, including an explicit inverse. Proposition 4.2 gives the resulting quotient.

2. **Constituents detected in the specified band generator.** A width-two band is an odd path of alternating pairs with fixed endpoint letters. At each position, a covector on the pair-letter space is represented by a two-by-two matrix. Pairing along the path evaluates a word of matrices and adjugates. To test a partition with at most four rows, place its tableau columns on consecutive path segments and alternate the covectors within each column. Each segment has length at most four. A four-element basis of $\text{Mat}_2(\mathbb{C})$ makes every such alternated segment matrix invertible. A simultaneous conjugation of the test basis makes the endpoint entry of the product nonzero. The resulting dual-polytabloid pairing proves that every shape with at most four rows occurs in $U_{m,2}$ itself (Proposition 5.3).
3. **A horizontal-strip reduction with four steps.** If a shape outside this dominance range has a first row of length at least m , one horizontal m -strip suffices. Otherwise its first four rows contain more than $2m - 1$ boxes. Successive sweeps of the bottom boxes of columns then remove exactly $2m - 1$ boxes in at most four horizontal strips (Lemma 6.1). In this second case, let ν be the remaining partition. Iterated Pieri supplies a shape $\eta \vdash 2m - 1$ with at most four rows such that S^λ occurs in the representation induced from $S^\nu \boxtimes S^\eta$. The band calculation puts S^η in $U_{m,2}$, while induction puts S^ν in W_{m-2} . The full induced quotient then gives the target constituent in W_m .

The smaller factor in the quotient is the actual cyclic module W_{m-s} , and the band factor is generated by the actual projected vector. This is why both parts of the induction track generators instead of using ambient tensor-product support alone. Only one copy of a suitable intermediate irreducible is needed.

Section 2 fixes the tensor conventions and proves the needed form of Young's rule from Pieri. Section 3 defines W_m and proves the dominated case. The quotient and band calculations occupy Sections 4 and 5; Section 6 proves the strip reduction and both main theorems.

2 Specht modules and horizontal strips

We fix the tensor conventions used to identify alternating block vectors with Specht modules. The other input needed by the induction is the Pieri rule: it transfers constituent support across the removal of horizontal strips. We record its dominance consequences with proofs.

All representations and tensor products are over $\mathbb{C}$ and are finite-dimensional. A partition $\lambda \vdash n$ is padded with zero parts when necessary; its length is $\text{len}(\lambda)$ and its conjugate is λ^t , where $\lambda_j^t = \#\{i : \lambda_i \geq j\}$. We identify λ with its diagram $\{(i, j) : 1 \leq j \leq \lambda_i\}$. For partitions of the same size, write $\lambda \supseteq \mu$ if $\sum_{i=1}^k \lambda_i \geq \sum_{i=1}^k \mu_i$ for every $k \geq 1$. Conjugation reverses this order:

$$\lambda \supseteq \mu \iff \mu^t \supseteq \lambda^t. \quad (2.1)$$

Indeed, for every integer $q \geq 1$,

$$\sum_i \max(\lambda_i - q, 0) = \max_{k \geq 0} \left(\sum_{i=1}^k \lambda_i - kq \right), \quad \sum_{j=1}^q \lambda_j^t = n - \sum_i \max(\lambda_i - q, 0).$$

Dominance increases the first expression and therefore decreases the second. Applying the resulting implication to the conjugate partitions proves the converse.

2.1 Position tensors and polytabloids

For a finite set B , write S_B for its symmetric group. For $\lambda \vdash |B|$, write S_B^λ for the irreducible S_B -module of type λ ; omit the subscript when the position set is understood. If E has ordered basis $e_1, \dots, e_d$, the tensor space $E^{\otimes B}$ has word basis $e_a = \bigotimes_{b \in B} e_{a(b)}$, indexed by maps $a : B \rightarrow \{1, \dots, d\}$. The position action is $ge_a = e_{a \circ g^{-1}}$. All regroupings of tensor factors are ordinary tensor identifications and introduce no signs.

Let L be a set partition of B whose blocks have size at most d . For each block $A = (b_1, \dots, b_r)$ choose an ordering and set

$$v_A = \sum_{\pi \in S_r} \text{sgn}(\pi) \bigotimes_{i=1}^r e_{\pi(i)}, \quad v_L = \bigotimes_{A \in L} v_A, \quad (2.2)$$

where the i th factor of v_A occupies position b_i . Changing a block ordering changes v_L only by a sign. Its nonzero word coefficients are ± 1 : on each block of size r , the letters are a bijection with $\{1, \dots, r\}$.

Lemma 2.1 (Alternating blocks as polytabloids). *Define $\theta_i = \#\{A \in L : |A| \geq i\}$, so that the block sizes are the column lengths of θ . Then $\mathbb{C}[S_B]v_L \cong S^\theta$. The same assertion holds for any ordered basis of a dual vector space.*

Proof. Arrange the blocks as columns of a tableau t of shape θ , in nonincreasing order of their sizes, and place each ordered block from top to bottom. Tableau entries are the distinct elements of B . A row tabloid $\{t'\}$ records the set of entries in each row. Let M^θ be the permutation module spanned by these row tabloids. Send $\{t'\}$ to the word assigning letter i to every entry of row i . This is an S_B -equivariant bijection from the tabloid basis of M^θ to the word basis of content θ . If C_t is the subgroup permuting entries within each column of t , the polytabloid

$$e_t = \sum_{c \in C_t} \text{sgn}(c) c\{t\}$$

maps to v_L : in a column, the position action replaces a permutation of the letters by its inverse, which has the same sign. In particular this vector is nonzero, since the original row word has coefficient 1. The span of all translates of e_t is the Specht module S^θ , irreducible over $\mathbb{C}$ by the standard polytabloid construction [Jam78, Definitions 4.1 and 4.3, Theorem 4.12]. This proves the assertion. The argument applies to a dual space with any chosen basis; alternatively, an invertible basis change on that space induces an invertible map on its position tensor power commuting with S_B . $\square$

We use the usual complex representation-theoretic facts [Jam78, Theorems 4.12 and 6.7]: the S^λ , for $\lambda \vdash n$, form the complete list of irreducible S_n -modules, and

$$(S^\lambda)^* \cong S^\lambda, \quad S^\lambda \otimes \text{sgn} \cong S^{\lambda^t}. \quad (2.3)$$

Every complex representation of a finite group is semisimple [FH91, Proposition 1.5 and Corollary 1.6], so any irreducible quotient is also a constituent. For $H \leq G$, Frobenius reciprocity [FH91, Proposition 3.17] takes the form

$$\text{Hom}_G(\text{Ind}_H^G A, V) \cong \text{Hom}_H(A, \text{Res}_H^G V). \quad (2.4)$$

Induction, modeled by $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} A$, is transitive and distributes over direct sums. The symbol $\boxtimes$ denotes the outer tensor product for separate group actions; ordinary tensor products of two S_B -modules carry the diagonal action. The natural pairing of $E^{\otimes B}$ and $(E^*)^{\otimes B}$ satisfies $\langle gu, gt \rangle = \langle u, t \rangle$. Thus pairing with an invariant subspace T of the dual tensor space defines an equivariant map $u \mapsto (t \mapsto \langle u, t \rangle)$ into T^* .

2.2 Pieri and the needed part of Young's rule

For diagrams $\nu \subseteq \lambda$, the skew diagram λ/ν is a *horizontal strip* if it has at most one box in each column. We use the empty partition and the trivial group S_0 in the following standard identity.

Lemma 2.2 (Pieri rule). *For $\nu \vdash a$ and $b \geq 0$,*

$$\text{Ind}_{S_a \times S_b}^{S_{a+b}}(S^\nu \boxtimes \mathbf{1}) \cong \bigoplus_{\substack{\lambda \vdash a+b, \lambda \supseteq \nu \\ \lambda/\nu \text{ a horizontal strip}}} S^\lambda. \quad (2.5)$$

Every summand occurs once.

This is the one-row specialization of the Littlewood–Richardson rule [Jam78, Section 16, especially Rule 16.4]: all b entries of its skew filling are 1, so column strictness requires a horizontal strip; the unique such filling satisfies the lattice-word condition. We give the dominance and multiplicity consequences needed here, including the constructive existence argument.

Lemma 2.3 (Young's rule: support and diagonal multiplicity). *Let $\theta = (\theta_1, \dots, \theta_\ell) \vdash n$. The row-tabloid permutation module has the equivalent induced realization*

$$M^\theta = \text{Ind}_{S_{\theta_1} \times \dots \times S_{\theta_\ell}}^{S_n} \mathbf{1}.$$

Then S^τ occurs in M^θ if and only if $\tau \supseteq \theta$, and S^θ occurs once. The word module of any content obtained by reordering the parts of θ is also isomorphic to M^θ .

Proof. The case $n = 0$ is immediate from the trivial representation of S_0 . Assume henceforth that $n > 0$. Transitivity of induction and Lemma 2.2 show that the multiplicity of S^τ counts chains

$$\emptyset = \nu^{(0)} \subseteq \nu^{(1)} \subseteq \dots \subseteq \nu^{(\ell)} = \tau, \quad |\nu^{(k)}/\nu^{(k-1)}| = \theta_k, \quad (2.6)$$

whose successive differences are horizontal strips. Each strip raises the first column's height by at most one, so $\nu^{(k)}$ has at most k rows. Since it is contained in τ ,

$$\sum_{i=1}^k \theta_i = |\nu^{(k)}| \leq \sum_{i=1}^k \tau_i.$$

For $k > \ell$, both initial sums equal n , since τ has at most ℓ rows. This proves necessity of dominance. If $\tau = \theta$, equality forces $\nu^{(k)}$ to be exactly the first k rows of θ . These diagrams do form a chain of the required kind, proving multiplicity one.

For sufficiency, suppose $\tau \supseteq \theta$ and induct on ℓ . For $\ell = 1$, necessarily $\tau = (n)$. For $\ell > 1$, put $t = \theta_\ell$. There are at least t columns of τ , since $\tau_1 \geq \theta_1 \geq t$. Delete a bottom box from each of the t tallest columns, choosing the rightmost columns first among equal heights, and call the remaining diagram ν . Its column heights are still nonincreasing: unequal adjacent heights differ by at least one, and within any group of equal heights precisely a suffix is lowered. Thus ν is a partition and τ/ν is a horizontal t -strip. For every $k \geq 1$ its loss in the first k rows is

$$\sum_{i=1}^k \tau_i - \sum_{i=1}^k \nu_i = \max(0, t - \tau_{k+1}) : \quad (2.7)$$

there are τ_{k+1} columns taller than k , and the rule selects these before any column whose bottom box belongs to the first k rows. If the loss is zero, the desired dominance inequality for ν follows immediately. If it is positive and $k < \ell$, then

$$\sum_{i=1}^k \nu_i = \sum_{i=1}^{k+1} \tau_i - t \geq \sum_{i=1}^{k+1} \theta_i - t \geq \sum_{i=1}^k \theta_i.$$

For completeness, dominance at ℓ gives $\text{len}(\tau) \leq \ell$, and dominance at $\ell - 1$ gives $\tau_\ell \leq t$. All τ_ℓ columns of height ℓ are therefore selected, so the last row disappears and $\text{len}(\nu) \leq \ell - 1$. Hence all remaining initial sums equal the total $n - t$, and $\nu \supseteq (\theta_1, \dots, \theta_{\ell-1})$. Induction gives a chain ending at ν ; appending the deleted strip gives Equation (2.6) ending at τ .

Finally, the stabilizer of a word with prescribed letter multiplicities is the product of the symmetric groups on its equal-letter position sets. Reordering those multiplicities conjugates this subgroup, giving the asserted isomorphism of permutation modules. $\square$

3 The triangular cyclic module and dominance

We now fix the generator from the introduction exactly and state the cyclic support theorem used in the induction. The second task of this section is to detect every constituent dominated by the staircase, using a sign twist and an explicit permutation-module quotient.

Following the triangular arrangement in [Ike15, Section 4], for $m \geq 1$ use the position set

$$B_m = \{(i, j) \in \mathbb{Z}_{\geq 1}^2 : i + j \leq m + 1\}, \quad G_m = S_{B_m}. \quad (3.1)$$

Its row and column set partitions are

$$\begin{aligned} R_m &= \{R_{m,i} : 1 \leq i \leq m\}, & R_{m,i} &= \{(i, j) : 1 \leq j \leq m + 1 - i\}, \\ C_m &= \{C_{m,j} : 1 \leq j \leq m\}, & C_{m,j} &= \{(i, j) : 1 \leq i \leq m + 1 - j\}. \end{aligned}$$

Order each row by increasing j and each column by increasing i . With $E_m = \mathbb{C}^m$ and its ordered basis $e_1, \dots, e_m$, form the alternating block tensors of Lemma 2.1 and set

$$w_m = v_{R_m} \otimes v_{C_m}, \quad W_m = \mathbb{C}[G_m]w_m \subseteq E_m^{\otimes B_m} \otimes E_m^{\otimes B_m}. \quad (3.2)$$

Both set partitions have block sizes $m, m - 1, \dots, 1$. Since $\rho_m^t = \rho_m$, Lemma 2.1 gives

$$V_R := \mathbb{C}[G_m]v_{R_m} \cong S^{\rho_m}, \quad V_C := \mathbb{C}[G_m]v_{C_m} \cong S^{\rho_m}, \quad W_m \subseteq V_R \otimes V_C. \quad (3.3)$$

Here and throughout, the action on a tensor product of two G_m -modules is diagonal. Thus it suffices to prove the following stronger statement.

Theorem 3.1 (Cyclic staircase support). *For every $m \geq 1$ and every partition $\lambda \vdash N_m$, the irreducible S^λ occurs in W_m .*

We prove Theorem 3.1 by induction in Section 6. The present section establishes the case supplied by dominance.

Proposition 3.2. *If $m \geq 1$, $\lambda \vdash N_m$, and $\rho_m \triangleright \lambda$, then S^λ occurs in W_m .*

Proof. Fix m , abbreviate $G = G_m$, and let ϵ be the one-dimensional sign representation, also denoting its character by ϵ . By Young's rule, conjugation of dominance, and the sign-twist identity in Equation (2.3), it is enough to show that $W_m \otimes \epsilon$ contains every constituent of M^{ρ_m} . We will find the cyclic module generated by $v_{R_m} \otimes v_{R_m}$ inside an isomorphic image of $W_m \otimes \epsilon$, then map that module onto M^{ρ_m} . The row subgroup

$$H_R = \prod_{i=1}^m S_{R_{m,i}}$$

acts on v_{R_m} by $\epsilon|_{H_R}$. Consequently the unnormalized signed diagonal average is exactly

$$\sum_{h \in H_R} \epsilon(h) h w_m = v_{R_m} \otimes u, \quad u = \sum_{h \in H_R} h v_{C_m}. \quad (3.4)$$

To see that $u \neq 0$, let $a_0(i, j) = i$ and consider the corresponding word e_{a_0} . It is fixed by H_R . Its coefficient in v_{C_m} is 1: within each increasingly ordered column it is the identity assignment $1, 2, \dots, m+1-j$. The coefficient of e_{a_0} in every $h v_{C_m}$ is therefore also 1, since $h^{-1} e_{a_0} = e_{a_0}$. Its coefficient in u is $|H_R| = \prod_{r=1}^m r!$, which is nonzero.

We next identify the alternating line in V_R . Frobenius reciprocity, the induction identity for a sign twist, and Lemma 2.3 give

$$\begin{aligned} \dim \operatorname{Hom}_{H_R}(\epsilon|_{H_R}, \operatorname{Res}_{H_R}^G V_R) &= \dim \operatorname{Hom}_G(\operatorname{Ind}_{H_R}^G \epsilon|_{H_R}, V_R) \\ &= \dim \operatorname{Hom}_G(M^{\rho_m} \otimes \epsilon, V_R) \\ &= \dim \operatorname{Hom}_G(M^{\rho_m}, V_R \otimes \epsilon) = 1. \end{aligned} \quad (3.5)$$

For the last equality, $V_R \otimes \epsilon \cong S^{\rho_m}$ by self-conjugacy of the staircase, and S^{ρ_m} has multiplicity one in M^{ρ_m} . As v_{R_m} is nonzero and alternating under H_R , it spans the line in Equation (3.5).

Twisting the diagonal tensor product can be done in its second factor: the reassociation

$$(V_R \otimes V_C) \otimes \epsilon \longrightarrow V_R \otimes (V_C \otimes \epsilon), \quad (x \otimes y) \otimes z \longmapsto x \otimes (y \otimes z) \quad (3.6)$$

is a G -isomorphism. Choose a G -isomorphism $\phi : V_C \otimes \epsilon \longrightarrow V_R$, which exists by Equation (3.3) and $\rho_m^t = \rho_m$, and choose $0 \neq z \in \epsilon$. Since u is H_R -fixed, $u \otimes z$ is alternating under H_R . It follows from Equation (3.5) that $\phi(u \otimes z) = c v_{R_m}$ for some $c \neq 0$. Tensor Equation (3.4) with z and apply the reassociation in Equation (3.6), followed by $\operatorname{id} \otimes \phi$. The right-hand side becomes $c(v_{R_m} \otimes v_{R_m})$. Thus the image of $W_m \otimes \epsilon$ under this isomorphism contains

$$Z_m = \mathbb{C}[G](v_{R_m} \otimes v_{R_m}) \subseteq V_R \otimes V_R. \quad (3.7)$$

We construct a surjection from Z_m onto M^{ρ_m} . Let E_{out} have basis $f_1, \dots, f_{2m-1}$ and define the linear map

$$q : E_m \otimes E_m \longrightarrow E_{\text{out}}, \quad q(e_a \otimes e_b) = f_{a+b-1}.$$

Regroup the two input layers by position, using ordinary tensor reassociation, and apply q at every position. Then project onto the span $\mathcal{M}_\alpha$ of words in which letter r occurs exactly r times for $1 \leq r \leq m$, and no letter greater than m occurs. Thus the content is $\alpha = (1, 2, \dots, m, 0, \dots, 0)$ in the output alphabet. Denote the resulting linear map by

$$Q : E_m^{\otimes B_m} \otimes E_m^{\otimes B_m} \longrightarrow \mathcal{M}_\alpha. \quad (3.8)$$

Both operations commute with position permutations: the same local map q is used everywhere, and content is unchanged by G . Hence Q is G -equivariant.

Consider any summand in the expansion of $v_{R_m} \otimes v_{R_m}$. On a row of length r , write its two assignments as $(a_1, \dots, a_r)$ and $(b_1, \dots, b_r)$, each a permutation of $1, \dots, r$, and put $c_j = a_j + b_j - 1$. Then

$$\sum_{j=1}^r c_j = r^2, \quad \sum_{j=1}^r c_j^2 - r^3 = \sum_{j=1}^r (c_j - r)^2 \geq 0. \quad (3.9)$$

For a term retained by the content projection, the sum of squared output letters over the whole triangle is $\sum_{r=1}^m r^3$. Summing Equation (3.9) over the rows forces equality on every row. Thus $c_j = r$ at every position in the row of length r . There is only one possible retained output word, namely

$$t_m = \bigotimes_{(i,j) \in B_m} f_{m+1-i}.$$

It remains to compute its coefficient. On a row of length r , each of the $r!$ assignments $a_j = \pi(j)$ has the unique compatible assignment $b_j = r + 1 - \pi(j)$. The latter is the permutation $\omega_r \pi$, where $\omega_r(j) = r + 1 - j$, so the product of the two alternating signs is

$$\text{sgn}(\pi) \text{sgn}(\omega_r \pi) = \text{sgn}(\omega_r) = (-1)^{\binom{r}{2}}.$$

The row contributions are independent. In particular, there is no cancellation, and the exact image is

$$Q(v_{R_m} \otimes v_{R_m}) = \left((-1)^{\sum_{r=1}^m \binom{r}{2}} \prod_{r=1}^m r! \right) t_m \neq 0. \quad (3.10)$$

The stabilizer of t_m is exactly H_R , since its equal-letter sets are the rows of B_m . Its orbit is a basis of $\mathcal{M}_\alpha$, and hence

$$Q(Z_m) = \mathbb{C}[G]t_m = \mathcal{M}_\alpha \cong \text{Ind}_{H_R}^G \mathbf{1} \cong M^{\rho_m}. \quad (3.11)$$

This also explains why the increasing content $(1, 2, \dots, m)$ gives the usual module indexed by the decreasing partition ρ_m : reordering the content parts conjugates the standard Young subgroup.

By Lemma 2.3, every S^τ with $\tau \triangleright \rho_m$ occurs in the quotient in Equation (3.11). Semisimplicity and Equation (3.7) imply that it occurs in $W_m \otimes \epsilon$. If $\rho_m \triangleright \lambda$, reversal of dominance under conjugation gives $\lambda^t \triangleright \rho_m^t = \rho_m$. Taking $\tau = \lambda^t$ and twisting back by ϵ proves that S^λ occurs in W_m . $\square$

4 A quotient obtained by cutting off a band

The next step transfers support from a smaller cyclic staircase module and a band into W_m . The projection must retain their specified generators and realize the full induced module, not just a submodule of it. We first define the two factors and then verify both requirements.

Fix $s \in \{1, 2\}$ and $h = m - s \geq 1$, and put

$$D = B_h, \quad F = B_m \setminus D, \quad G = G_m = S_{B_m}, \quad H = S_D \times S_F. \quad (4.1)$$

Thus $|F| = N_m - N_h = s(2m - s + 1)/2$. Let R_F and C_F be the partitions of F into its nonempty row and column intersections, with the orders inherited from R_m and C_m . Each block has size at most s . In two layers with alphabet $1, \dots, s$, define

$$u_{m,s} = v_{R_F} \otimes v_{C_F}, \quad U_{m,s} = \mathbb{C}[S_F]u_{m,s}. \quad (4.2)$$

The dependence on m records the length of the band. Our goal is a quotient of W_m induced from $W_h \boxtimes U_{m,s}$.

In the disjoint-coset model of induction [FH91, Section 3.3], translates of the inducing module form a direct sum. We record why disjoint coordinate sectors realize this model for the cyclic span of a specified vector, including the inverse map.

Lemma 4.1 (Induction from coordinate sectors). *Let a finite group G act transitively on a finite set Ω . Suppose a G -module has a direct-sum decomposition $T = \bigoplus_{A \in \Omega} T_A$ with $gT_A = T_{gA}$. Fix $D \in \Omega$, let H be its stabilizer, and take $z \in T_D$. If $A_0 = \mathbb{C}[H]z$, then the map*

$$\Phi : \mathbb{C}[G] \otimes_{\mathbb{C}[H]} A_0 \longrightarrow \mathbb{C}[G]z, \quad g \otimes a \longmapsto ga \quad (4.3)$$

is an isomorphism of G -modules.

Proof. The relation $(gh) \otimes a = g \otimes ha$, for $h \in H$, has the same image on both sides, so Φ is well-defined. It is equivariant and surjective. Choose $g_A \in G$ with $g_AD = A$ for every $A \in \Omega$, and let $\pi_A : T \rightarrow T_A$ be the coordinate projection. Since

$$\mathbb{C}[G]z = \bigoplus_{A \in \Omega} g_A A_0, \quad g_A A_0 \subseteq T_A,$$

we have $g_A^{-1}\pi_A(y) \in A_0$ for $y \in \mathbb{C}[G]z$. Define

$$\Psi(y) = \sum_{A \in \Omega} g_A \otimes g_A^{-1}\pi_A(y). \quad (4.4)$$

Replacing g_A by $g_A h$ does not change this expression, because $g_A h \otimes h^{-1}a = g_A \otimes a$ in the balanced tensor product. Clearly $\Phi\Psi(y) = \sum_A \pi_A(y) = y$. For $g \otimes a$, set $A = gD$ and write $g = g_A h$ with $h \in H$. Only its A -sector is nonzero, and hence

$$\Psi\Phi(g \otimes a) = g_A \otimes ha = g \otimes a.$$

Thus the displayed maps are inverse isomorphisms. $\square$

Proposition 4.2 (Band quotient). *For the parameters in Equation (4.1), there is a surjective G_m -equivariant map*

$$W_m \twoheadrightarrow \text{Ind}_{S_D \times S_F}^{S_{B_m}}(W_h \boxtimes U_{m,s}). \quad (4.5)$$

In particular, every irreducible constituent of the induced module on the right is a constituent of W_m .

Proof. Projection and the high positions. Use the ordinary tensor identification

$$E_m^{\otimes B_m} \otimes E_m^{\otimes B_m} \cong (E_m \otimes E_m)^{\otimes B_m}.$$

Put $E_{\text{lo}} = \text{span}(e_1, \dots, e_s)$ and $E_{\text{hi}} = \text{span}(e_{s+1}, \dots, e_m)$. On a pair-letter basis vector define

$$p_s(e_a \otimes e_b) = \begin{cases} e_a \otimes e_b, & a, b \leq s \text{ or } a, b > s, \\ 0, & \text{otherwise,} \end{cases} \quad P_s = p_s^{\otimes B_m}.$$

The same local projection is used at every position, so P_s commutes with G . Call a retained position *high* when both letters are greater than s .

In a retained word of w_m , let A be the set of high positions. Alternation in the first layer gives row counts $r_i = \max(h+1-i, 0)$, and alternation in the second gives column counts $d_j = \max(h+1-j, 0)$, for $1 \leq i, j \leq m$. For each $1 \leq k \leq m$, these counts satisfy

$$\sum_{i=1}^k r_i = \#\{(i, j) \in B_h : i \leq k\} = \sum_{j=1}^m \min(k, d_j). \quad (4.6)$$

Column j can contribute at most $\min(k, d_j)$ high positions to the first k rows. Equality of the sums in Equation (4.6) therefore forces equality in every column. For $d_j > 0$, take $k = d_j$: all its d_j high positions occupy the first d_j rows. Columns with $d_j = 0$ have no high positions. Consequently

$$A = \{(i, j) \in B_m : i \leq h+1-j\} = B_h = D. \quad (4.7)$$

For the zero-one matrix recording membership in A , this is the equality case of the row and column margin inequalities [Rys57, Section 2]. The specific staircase margins in Equation (4.6) make every prefix bound an equality.

Factorization with signs. In a row or column block of length r , its intersection with D is its initial segment of length $a = \max(r-s, 0)$; its intersection with F has length $b = \min(r, s)$. The retained assignments put the high letters $s+1, \dots, r$ on the first segment and the low letters $1, \dots, b$ on the second. Let Alt denote the unnormalized alternation of a displayed list, placed in the indicated segment, with $\text{Alt}(\emptyset) = 1$. The part of the block tensor with this prescribed high set is exactly

$$(-1)^{ab} \text{Alt}(e_{s+1}, \dots, e_r) \otimes \text{Alt}(e_1, \dots, e_b). \quad (4.8)$$

Here the high list is empty when $r \leq s$. Indeed, every high letter precedes every low letter in the position order, giving ab inversions; all other inversions are internal to the two independently permuted lists. Every such pair of permutations occurs once. In particular, when $r < s$ the formula is just the original alternation on $e_1, \dots, e_r$.

Rows and columns both have the length list $m, m-1, \dots, 1$. Thus their cross signs in Equation (4.8) multiply to

$$\prod_{r=1}^m (-1)^{2 \max(r-s, 0) \min(r, s)} = 1.$$

Identify E_h with E_{hi} by $e_a \mapsto e_{s+a}$, and write $\tilde{w}_h$ and $\widetilde{W}_h$ for the images of w_h and W_h in the two high-letter layers on D . The low portions are precisely the blocks of R_F, C_F . Regrouping the factors over the disjoint sets D, F introduces no signs, and hence

$$z := P_s(w_m) = \tilde{w}_h \otimes u_{m,s} \neq 0. \quad (4.9)$$

The nonvanishing follows also directly from the nonzero alternating tensors on each disjoint block.

The two factors of $H = S_D \times S_F$ act independently on this tensor, while each acts diagonally on the two layers over its own position set. Therefore

$$\begin{aligned} A_0 := \mathbb{C}[H]z &= \text{span}\{(\sigma \tilde{w}_h) \otimes (\tau u_{m,s}) : \sigma \in S_D, \tau \in S_F\} \\ &= \widetilde{W}_h \boxtimes U_{m,s} \cong W_h \boxtimes U_{m,s}. \end{aligned} \quad (4.10)$$

The second equality follows by expanding tensors of linear combinations of the two separate orbit sets.

The coordinate sectors. Let Ω be the set of subsets $A \subset B_m$ of size $|D|$. Within the pair-letter space define

$$T_A = (E_{\text{hi}} \otimes E_{\text{hi}})^{\otimes A} \otimes (E_{\text{lo}} \otimes E_{\text{lo}})^{\otimes (B_m \setminus A)}.$$

These are the coordinate spans of words high exactly on A . They have disjoint word bases, so their sum is direct, and $gT_A = T_{gA}$. By Equation (4.9), $z \in T_D$; the stabilizer of D in G is exactly H . Apply Lemma 4.1 and Equation (4.10) to obtain

$$\mathbb{C}[G]z \cong \text{Ind}_H^G A_0 \cong \text{Ind}_H^G (W_h \boxtimes U_{m,s}).$$

Since P_s is equivariant, $P_s(W_m) = \mathbb{C}[G]P_s(w_m) = \mathbb{C}[G]z$. Its restriction to W_m , followed by the inverse map Ψ in Equation (4.4) and the identification of A_0 in Equation (4.10), gives Equation (4.5). The image of this ambient projection need not lie in the two original Specht factors; the composition is still a surjective G -equivariant linear map onto the displayed module. Semisimplicity over $\mathbb{C}$ proves the last assertion. $\square$

5 Cyclic support of the bands

For the band step of the induction, it remains to determine the support needed in $U_{m,1}$ and $U_{m,2}$ for Proposition 4.2. The width-one factor is trivial. For width two we will detect every shape with at most four rows directly in the specified generator, using the invariant evaluation pairing.

For width one, the band consists of the m positions with $i + j = m + 1$. Every row and column block is a singleton, and both alphabets consist only of the letter 1. Thus $u_{m,1}$ is a nonzero constant word tensor, fixed by every position permutation, and

$$U_{m,1} \cong S^{(m)} \quad (m \geq 2). \quad (5.1)$$

We now determine the support needed from the particular width-two generator.

5.1 The indexed path contraction

Let $m \geq 3$, $K = 2m - 1$, and $F = B_m \setminus B_{m-2}$. Number its positions along the path by

$$p_{2j-1} = (m + 1 - j, j) \quad (1 \leq j \leq m), \quad p_{2j} = (m - j, j) \quad (1 \leq j < m). \quad (5.2)$$

In this numbering, the row blocks are the singleton 1 and the pairs $(2, 3), (4, 5), \dots, (K - 1, K)$; the column blocks are $(1, 2), (3, 4), \dots, (K - 2, K - 1)$ and the singleton K . Order each pair by increasing path index and let u denote the resulting alternating band tensor. The row orders agree with increasing j , while each of the $m - 1$ column pairs reverses increasing i . Consequently

$$u = (-1)^{m-1} u_{m,2}, \quad U_{m,2} = \mathbb{C}[S_F]u. \quad (5.3)$$

Figure 1 illustrates the numbering and the consecutive segments used in Example 5.4.

Put $E = \mathbb{C}^2$ with basis e_1, e_2 and $V = E \otimes E$. Ordinary tensor reassociation identifies the two band layers with $V^{\otimes K}$, using the position order in Equation (5.2). Identify V^* with $\text{Mat}_2(\mathbb{C})$ by letting a matrix X take the value X_{ab} on $e_a \otimes e_b$. All pairings below are bilinear evaluation pairings. Write

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

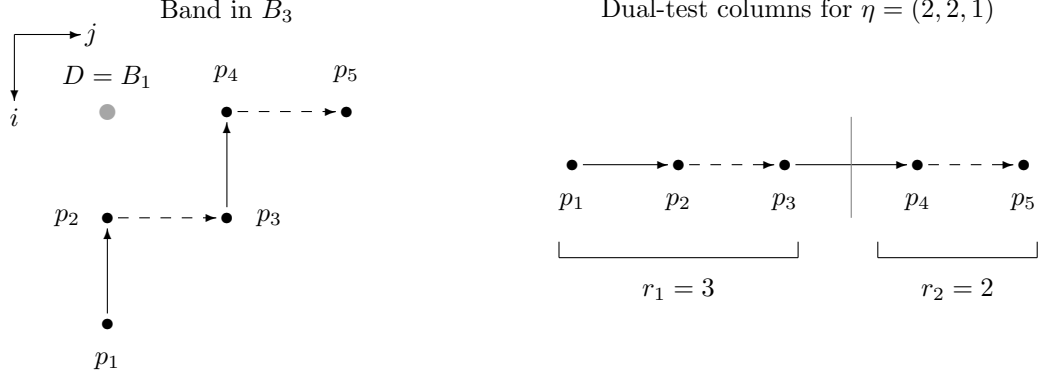


Figure 1: The width-two band for $m = 3$. Solid arrows are column pairs; dashed arrows are row pairs, all directed by increasing path index. Column arrows reverse increasing i , as in Equation (5.3). The row singleton is p_1 and the column singleton is p_5 , each carrying the fixed letter 1 in its layer. On the right, the dual-test tableau of shape $\eta = (2, 2, 1)$ has column lengths $r_1 = 3$ and $r_2 = 2$, giving the two bracketed consecutive segments. The gray vertical line separates these segments, which differ from the original pair blocks.

Lemma 5.1 (Path contraction). *For arbitrary $X_1, \dots, X_K \in \text{Mat}_2(\mathbb{C})$,*

$$\langle u, X_1 \otimes \dots \otimes X_K \rangle = (-1)^{m-1} [X_1 \text{adj}(X_2) X_3 \text{adj}(X_4) \dots X_{K-2} \text{adj}(X_{K-1}) X_K]_{11}. \quad (5.4)$$

Proof. The two singleton blocks require the first row-layer index and last column-layer index to be 1. The coefficient of an alternating pair in path order is J_{ab} . Hence the left side of Equation (5.4) is exactly

$$\sum_{\substack{a_1, \dots, a_K, b_1, \dots, b_K \in \{1, 2\} \\ a_1 = b_K = 1}} \left(\prod_{\ell=1}^{m-1} J_{b_{2\ell-1}, b_{2\ell}} J_{a_{2\ell}, a_{2\ell+1}} \right) \prod_{q=1}^K (X_q)_{a_q b_q}. \quad (5.5)$$

Reading the contracted indices successively from a_1 to b_K gives

$$\left[X_1 J X_2^\top J X_3 J X_4^\top J \dots X_{K-2} J X_{K-1}^\top J X_K \right]_{11}.$$

The transpose at an even position records that its column index is encountered before its row index. For $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, direct multiplication gives

$$\text{adj}(X) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \text{tr}(X)I - X, \quad JX^\top J = -\text{adj}(X). \quad (5.6)$$

There are $m - 1$ even positions, yielding the stated sign. In particular, adjugation is a linear operation on $\text{Mat}_2(\mathbb{C})$. $\square$

5.2 Alternating consecutive segments

For a shape with at most four rows, place its tableau columns on successive disjoint segments of the path. On a segment of length r , the dual polytabloid alternates the first r elements of one common basis of V^* . The contraction formula therefore assigns an alternating matrix sum to each segment;

these sums multiply in path order. We now choose the basis so that every such segment matrix, for $1 \leq r \leq 4$ and either starting parity, is invertible.

Use the following ordered basis of V^* :

$$M_1 = I, \quad M_2 = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad M_3 = T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_4 = J = ZT. \quad (5.7)$$

These matrices are linearly independent: I, Z span the diagonal matrices and T, J span the off-diagonal matrices. The last three are traceless and anticommute in pairs; moreover

$$Z^2 = T^2 = I, \quad J^2 = -I.$$

Thus all four are invertible. Alternating evaluations on Z, T, J are also used to test polynomial identities for two-by-two matrices with adjugation [DG94, Section 3]; here the adjugations are determined by the path positions.

For a global path position q , define the linear operation

$$D_q(X) = \begin{cases} X, & q \text{ odd}, \\ \text{adj}(X), & q \text{ even}. \end{cases}$$

For a segment of length $r \leq 4$ beginning at position p , define its alternating matrix by the following sum of ordered products:

$$A_{r,p} = \sum_{\pi \in S_r} \text{sgn}(\pi) D_p(M_{\pi(1)}) D_{p+1}(M_{\pi(2)}) \cdots D_{p+r-1}(M_{\pi(r)}). \quad (5.8)$$

Lemma 5.2 (Segment alternation). *Let e be the number of even positions among $p, \dots, p+r-1$, and put $\delta = 1$ if p is even and $\delta = 0$ otherwise. Then*

$$A_{r,p} = r!(-1)^{e-\delta} M_1 \cdots M_r. \quad (5.9)$$

In particular, all these matrices are invertible. Their values are

r	$p \text{ odd}$	$p \text{ even}$
1	I	I
2	$-2Z$	$2Z$
3	$-6J$	$-6J$
4	$-24I$	$24I$

(5.10)

Proof. Fix a permutation π and let j be the local slot occupied by $M_1 = I$. Deleting I leaves a permutation of $M_2, \dots, M_r$. Its sign is exactly the sign acquired when those pairwise anticommuting matrices are reordered. Moving I from its first canonical slot to slot j contributes $(-1)^{j-1}$ to $\text{sgn}(\pi)$ and changes no matrix product. It follows that

$$\text{sgn}(\pi) M_{\pi(1)} \cdots M_{\pi(r)} = (-1)^{j-1} M_1 \cdots M_r.$$

By Equation (5.6), adjugation fixes I and negates each of the other basis matrices. Write $\epsilon_j = 1$ if $p+j-1$ is even, and 0 otherwise. Adjugation therefore contributes the additional sign $(-1)^{e-\epsilon_j}$. Since $\epsilon_j \equiv \delta + j - 1 \pmod{2}$, the total sign is $(-1)^{e-\delta}$, independently of π . All $r!$ summands in Equation (5.8) consequently agree, proving Equation (5.9), including $r = 1$. Finally, the canonical products for $r = 1, 2, 3, 4$ are respectively $I, Z, J, -I$, which gives Equation (5.10). $\square$

5.3 Detection by a dual polytabloid

The segment calculation gives an invertible ordered product. Its fixed endpoint entry can nevertheless vanish. We now choose the dual test basis so that this entry is nonzero, while keeping the band generator and its endpoint letters unchanged.

Proposition 5.3 (Four-row support of the width-two band). *For every $m \geq 3$ and every partition $\eta \vdash 2m - 1$ with $\text{len}(\eta) \leq 4$, the irreducible module S^η occurs in $U_{m,2}$.*

Proof. Let $r_1, \dots, r_c$ be the column lengths of η in their usual order, so $1 \leq r_j \leq 4$ and $\sum_j r_j = K$. Partition the path into consecutive segments of these lengths, beginning at $s_j = 1 + \sum_{i < j} r_i$. For any ordered basis $N_1, \dots, N_4$ of V^* , let t_N be the dual tensor whose factor on segment j is

$$\sum_{\pi \in S_{r_j}} \text{sgn}(\pi) N_{\pi(1)} \otimes \cdots \otimes N_{\pi(r_j)}.$$

The segments are the columns of a tableau of shape η and the initial basis entries label its rows. By Lemma 2.1, their product t_N is a polytabloid, and

$$\mathcal{T}_N = \mathbb{C}[S_F]t_N \subseteq (V^*)^{\otimes K} \quad \text{is irreducible of type } \eta.$$

First take $N_i = M_i$. Multilinearity of Equation (5.4) and the definition of t_M give

$$\langle u, t_M \rangle = (-1)^{m-1} [B]_{11}, \quad B = A_{r_1, s_1} A_{r_2, s_2} \cdots A_{r_c, s_c}. \quad (5.11)$$

Indeed, summing independently over the permutations within each segment is exactly the distributive expansion of the displayed ordered product. The segments are consecutive, so no matrix factors are interchanged in this factorization. By Lemma 5.2, B is invertible.

The identity $\text{adj}(X) = \text{tr}(X)I - X$ also shows that, for every $Q \in \text{GL}_2(\mathbb{C})$,

$$D_q(QXQ^{-1}) = QD_q(X)Q^{-1}.$$

Choose an eigenvector v of B with eigenvalue β ; invertibility gives $\beta \neq 0$. Choose Q taking v to the first coordinate vector and set $N_i = QM_iQ^{-1}$. Conjugation is an invertible linear operation on V^* , so these N_i are still an ordered basis. Each alternating segment matrix is now $QA_{r_j, s_j}Q^{-1}$, and their ordered product is QBQ^{-1} . Since $(QBQ^{-1})e_1 = \beta e_1$,

$$\langle u, t_N \rangle = (-1)^{m-1} [QBQ^{-1}]_{11} = (-1)^{m-1} \beta \neq 0. \quad (5.12)$$

Only the test covectors have changed; u and its endpoint letters remain those fixed in Equations (5.3) and (5.5).

The invariant evaluation pairing defines

$$\Phi : U_{m,2} \longrightarrow \mathcal{T}_N^*, \quad \Phi(x)(t) = \langle x, t \rangle. \quad (5.13)$$

For $g \in S_F$, its equivariance follows explicitly from $\Phi(gx)(t) = \langle x, g^{-1}t \rangle = (g\Phi(x))(t)$. By Equation (5.12) this map is nonzero on the particular generator u . Its target is irreducible and, by the self-duality in Equation (2.3), has type η . Thus Φ is surjective, and semisimplicity proves the claimed occurrence in $U_{m,2}$. $\square$

Example 5.4. The change of test basis can be essential. For $m = 3$ and $\eta = (2, 2, 1)$, the column segments have lengths 3, 2, beginning at positions 1, 4. Their product in Equation (5.11) is $(-6J)(2Z) = 12T$, whose $(1, 1)$ entry vanishes. With

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad QTQ^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & -1 \end{pmatrix},$$

the resulting pairing is 12, since $(-1)^{m-1} = 1$.

6 Completion of the induction

We now combine the two band quotients with the dominance case. The combinatorial point is that a partition outside the dominance range either has a long enough first row for a width-one cut, or has enough boxes in its first four rows for a width-two cut.

Lemma 6.1 (Horizontal-strip reduction). *Let $m \geq 2$ and $\lambda \vdash N_m$. If λ is not dominated by ρ_m , then at least one of the following holds.*

- (i) *There is a partition $\nu \vdash N_{m-1}$ such that λ/ν is a horizontal strip of size m .*
- (ii) *One has $m \geq 5$, and there are partitions*

$$\nu = \lambda^{(0)} \subset \lambda^{(1)} \subset \cdots \subset \lambda^{(q)} = \lambda, \quad q \leq 4,$$

with $\nu \vdash N_{m-2}$ and every $\lambda^{(i)}/\lambda^{(i-1)}$ a nonempty horizontal strip. Their total size is $2m - 1$.

Proof. Unlike the deletion in Lemma 2.3, this reduction need not preserve a prescribed dominance inequality; a rightmost suffix suffices to preserve the partition. If $\lambda_1 \geq m$, remove the bottom box in each of the rightmost m nonempty columns. Subtracting one from this suffix of the column-height sequence preserves weak decrease and nonnegativity. The remaining boxes therefore form a partition ν , and the deleted boxes form a horizontal strip. Since $N_m - m = N_{m-1}$, this proves (i).

Suppose instead that $\lambda_1 \leq m - 1$. Failure of domination gives an index $k < m$ with

$$\sum_{i=1}^k \lambda_i > km - \frac{k(k-1)}{2}. \quad (6.1)$$

The index is less than m because the first m parts of the staircase already have sum N_m . For $k \leq 3$, however,

$$\sum_{i=1}^k \lambda_i \leq k(m-1) \leq km - \frac{k(k-1)}{2}.$$

Thus $4 \leq k < m$, and in particular $m \geq 5$. Because the parts decrease, the mean of the first four is at least the mean of the first k . Hence

$$\sum_{i=1}^4 \lambda_i \geq \frac{4}{k} \sum_{i=1}^k \lambda_i > 4m - 2(k-1) \geq 2m + 4 > 2m - 1. \quad (6.2)$$

A full sweep removes the bottom box of every nonempty column. If the current row lengths are $(\alpha_1, \alpha_2, \dots)$, subtracting one from all its positive column heights gives row lengths $(\alpha_2, \alpha_3, \dots)$. The remaining diagram is nested in the original one, and the removed cells form a horizontal strip of size α_1 . Four full sweeps starting from λ would therefore remove $\lambda_1 + \cdots + \lambda_4$ boxes.

Perform full sweeps until the remaining number of boxes to be removed is at most the number of nonempty columns, then remove that number of bottom boxes from the rightmost columns. The suffix argument in the first paragraph shows that this last partial sweep also leaves a partition. By Equation (6.2), at most four nonempty sweeps remove exactly $2m - 1$ boxes. Their remainder has size $N_m - (2m - 1) = N_{m-2}$. Reversing the sequence gives (ii). $\square$

Four strips can be necessary. Take $m = 13$ and $\lambda = (8^{11}, 3)$, where the exponent denotes eleven repeated parts equal to 8. This partition has size $91 = N_{13}$ and fails domination at $k = 12$, since its first twelve parts sum to $91 > 90$. Every horizontal strip in its eight-column diagram has at most eight boxes, so three strips cannot remove the required $25 = 2m - 1$ boxes. The sweeps remove $8 + 8 + 8 + 1 = 25$ boxes and leave $\nu = (8^7, 7, 3)$ of size $66 = N_{11}$.

Proof of Theorem 3.1. We induct on m . The module W_1 is the one-dimensional representation of the one-element group, so the assertion holds at $m = 1$.

Let $m \geq 2$, suppose the assertion holds for all smaller positive indices, and fix $\lambda \vdash N_m$. If $\rho_m \supseteq \lambda$, then Proposition 3.2 supplies S^λ in W_m . Otherwise apply Lemma 6.1.

In case (i), use the band cut with $s = 1$, so $|D| = N_{m-1}$ and $|F| = m$. The width-one band module $U_{m,1}$ is trivial. By the induction hypothesis, S_D^ν is a constituent of W_{m-1} . By Lemma 2.2, $S_{B_m}^\lambda$ occurs in

$$\text{Ind}_{S_D \times S_F}^{S_{B_m}}(S_D^\nu \boxtimes \mathbf{1}).$$

Consequently it occurs in the quotient from Proposition 4.2 and hence in W_m .

In case (ii), put $K = 2m - 1$ and use the cut with $s = 2$. Write $b_i = |\lambda^{(i)}| - |\lambda^{(i-1)}|$, so that $b_1 + \dots + b_q = K$ and $q \leq 4$. Choose disjoint subsets of F of these sizes and form the permutation module

$$A = \text{Ind}_{S_{b_1} \times \dots \times S_{b_q}}^{S_F} \mathbf{1} \cong \bigoplus_{\eta \vdash K} (S_F^\eta)^{\oplus a_\eta}.$$

Repeated application of Lemma 2.2, followed by transitivity of induction, shows that $S_{B_m}^\lambda$ is a constituent of $\text{Ind}_{S_D \times S_F}^{S_{B_m}}(S_D^\nu \boxtimes A)$. Equivalently,

$$\begin{aligned} \sum_{\eta \vdash K} a_\eta c_{\nu, \eta}^\lambda &> 0, \\ c_{\nu, \eta}^\lambda &= \dim \text{Hom}_{S_{B_m}} \left(S_{B_m}^\lambda, \text{Ind}_{S_D \times S_F}^{S_{B_m}}(S_D^\nu \boxtimes S_F^\eta) \right). \end{aligned} \tag{6.3}$$

Here the coefficients are nonnegative integers. Moreover, every constituent of A has at most q rows: starting from the empty diagram, each of the q horizontal-strip additions can introduce at most one new row, because two new rows would both require a new box in the first column. Thus Equation (6.3) provides some η with at most four rows, $a_\eta > 0$, and $c_{\nu, \eta}^\lambda > 0$.

By Proposition 5.3, this S_F^η occurs in $U_{m,2}$. By induction, S_D^ν occurs in the specified cyclic module W_{m-2} . One copy of each suffices to give $S_{B_m}^\lambda$ in

$$\text{Ind}_{S_D \times S_F}^{S_{B_m}}(W_{m-2} \boxtimes U_{m,2}),$$

which is a quotient of W_m by Proposition 4.2. Complete reducibility again gives the constituent in W_m itself. This completes the induction. Notice that case (ii) has $m \geq 5$, so every smaller module invoked has a positive index. $\square$

Proof of Theorem 1.1. The two factors generating W_m have Specht type ρ_m . Thus W_m is a submodule of $S^{\rho_m} \otimes_{\mathbb{C}} S^{\rho_m}$ under the diagonal action. Theorem 3.1 supplies every S^λ in that submodule, so $g(\rho_m, \rho_m, \lambda) > 0$ for every permitted m and λ . $\square$

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